

A PROJECT TITLE
ON
SPAM EMAIL CLASSIFICATION USING NAIVE BAYES AND SUPPORT
VECTOR MACHINE

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

K. Pranav
Abhay Tripathi

Roll No: 2520030477
Roll No: 3520030353

Under the Esteemed guidance of

Ms. Jyothsna Datti

<Designation, Department>

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Name of Title: Spam Email Classification Using Naive Bayes and Support Vector Machine

1. Abstract:

Spam Email Classification Using Naive Bayes and Support Vector Machine algorithms addresses the critical challenge of electronic communication security. As unwanted, promotional, and fraudulent emails continue to flood user inboxes, automating the detection of spam messages becomes imperative for maintaining productivity and safeguarding personal data. The primary objective of this project is to develop and evaluate an automated classification framework capable of distinguishing legitimate emails (ham) from unsolicited emails (spam). For this study, the SMS Spam Collection / Enron Email Dataset containing labeled text instances will be utilized for training and testing. Text pre-processing techniques including tokenization, stop-word removal, and TF-IDF feature extraction will be implemented before passing the vector representation into Multinomial Naive Bayes and Support Vector Machine (SVM) models. At the conclusion of the project, performance will be evaluated and presented using metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The expected outcome is a comparative evaluation showcasing which classifier achieves higher classification precision with minimal false positives.

2. Introduction:

The proliferation of unsolicited bulk electronic messages poses significant challenges to digital communication efficiency, storage management, and cyber security. Modern email users are subjected to a continuous stream of phishing attempts, advertisement clutter, and malicious links embedded within spam messages. This creates a compelling motivation to engineer intelligent automated systems that can analyze incoming text streams in real-time, filter out harmful content, and ensure seamless delivery of legitimate communications without relying on manual intervention.

Machine learning offers robust probabilistic and margin-based solutions to automate text categorization tasks. The Naive Bayes algorithm applies Bayes' theorem with strong independence assumptions, providing rapid computational training ideal for text feature spaces. Conversely, Support Vector Machines project high-dimensional text feature vectors onto optimal hyperplanes to separate classes effectively. By systematically implementing and benchmarking both Naive Bayes and Support Vector Machine models on standardized text data, this project provides insights into how probabilistic models compare against linear and non-linear kernel boundaries for text-based spam detection.

3. Software Requirements:

To execute this machine learning pipeline, specific software dependencies and hardware capabilities are required to handle dataset loading, natural language pre-processing, vectorization, hyperparameter tuning, model training, and performance evaluation efficiently. The setup ensures smooth experimentation with Python-based computational libraries and standard multi-core hardware execution environments.

Component Type	Requirement / Specification
Operating System	Windows 10/11, macOS, or Ubuntu Linux
Programming Language	Python 3.8+
Libraries & Frameworks	Scikit-Learn, Pandas, NumPy, NLTK, Matplotlib, Seaborn
Development Environment	Jupyter Notebook / VS Code / Google Colab
Processor (CPU)	Intel Core i5 / AMD Ryzen 5 or higher
RAM	8 GB minimum (16 GB recommended)
Storage	10 GB available SSD space

Signature of the Guide

Course Instructor